A PROJECT TITLE

ON

Credit Card Fraud Detection Using Logistic Regression, SVM and Isolation Forest

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

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1. **Abstract:**

Credit card fraud is a major financial security problem that causes significant monetary losses for customers, banks, and financial institutions. With the rapid growth of digital payments and online transactions, detecting fraudulent activities accurately and at an early stage has become increasingly important. This project, "Credit Card Fraud Detection Using Logistic Regression, SVM and Isolation Forest," focuses on developing a machine learning-based approach for identifying suspicious credit card transactions.

The project uses historical transaction data containing legitimate and fraudulent transactions to train and evaluate different machine learning techniques. Logistic Regression is used as a supervised classification algorithm to estimate the probability that a transaction is fraudulent. Support Vector Machine (SVM) is applied to identify an optimal decision boundary between legitimate and fraudulent transactions, making it suitable for complex classification problems. Isolation Forest is used as an unsupervised anomaly-detection technique to identify unusual transactions that differ significantly from normal transaction patterns.

The dataset is first preprocessed by handling missing values, selecting relevant features, scaling numerical attributes, and addressing the highly imbalanced nature of fraud datasets. The models are trained and tested using appropriate evaluation techniques. Their performance can be compared using metrics such as accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC. Since fraud detection requires minimizing missed fraudulent transactions, particular importance is given to precision and recall rather than accuracy alone.

The proposed project demonstrates how different machine learning approaches can be used to detect fraudulent credit card transactions. Comparing supervised and unsupervised techniques provides a better understanding of their strengths and limitations and can help identify an effective approach for improving financial transaction security.

2. INTRODUCTION:

Credit card transactions have become an essential part of modern financial activities because they provide fast, convenient, and secure methods of payment. However, the increasing volume of electronic transactions has also resulted in an increase in fraudulent activities. Credit card fraud occurs when a card or its information is used without the authorization of the legitimate cardholder. Such activities can result in financial losses and can reduce customer trust in digital payment services.

Traditional fraud detection methods generally depend on predefined rules and manually identified patterns. Although these methods can detect known types of fraud, they may have difficulty identifying new or continuously changing fraudulent behaviors. Machine learning provides an effective alternative because it can learn patterns from historical transaction data and use those patterns to identify potentially fraudulent transactions.

This project focuses on three machine learning techniques: **Logistic Regression, Support Vector Machine (SVM), and Isolation Forest**. Logistic Regression and SVM are supervised learning algorithms that learn from labeled transaction data. Logistic Regression predicts the probability of a transaction belonging to the fraudulent class, while SVM attempts to find an optimal boundary that separates legitimate and fraudulent transactions. Isolation Forest is an unsupervised anomaly detection algorithm that identifies transactions that are unusual compared with normal transaction behavior.

The main objective of this project is to build, evaluate, and compare these three approaches and determine how effectively they can identify fraudulent transactions while reducing incorrect classifications.

3. SOFTWARE REQUIREMENTS:

Software / Tool Requirement

Operating System:	Windows 10/11, Linux, or macOS
Programming Language:	Python 3.x
Development Environment:	Jupyter Notebook, Google Colab, or VS Code
Data Processing:	Pandas, NumPy
Machine Learning:	Scikit-learn
Data Visualization:	Matplotlib, Seaborn
Dataset:	Credit Card Fraud Detection dataset in CSV format
Model Algorithms:	Logistic Regression, SVM, Isolation Forest
Evaluation Metrics:	Accuracy, Precision, Recall, F1-score, ROC-AUC, Confusion Matrix

Signature of the Guide

Course Instructor